

Accurate singular values *and* singular vectors from mixed-precision Jacobi algorithms

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Small singular values get lost

Let $A \in \mathbb{R}^{m \times n}$ have full rank, $m \geq n$, with SVD $A = U\Sigma V^T$. Methods built on an initial bidiagonalization (MATLAB `svd`, `xGESVD`, `xGESDD`) satisfy only

$$\varepsilon_{\text{fwd}}^{(k)} := \frac{|\widehat{\sigma}_k - \sigma_k|}{\sigma_k} \leq p(m, n) u \kappa_2(A).$$

For an ill-conditioned A , the small singular values come out with *no relative accuracy at all*.

Demmel and Veselić [DV92] showed the one-sided Jacobi algorithm does better: its bound involves the **scaled** condition number

$$\kappa_2^S(A) = \kappa_2(AD), \quad D = \text{diag}(\|a_i\|_2^{-1}),$$

which can be significantly smaller than $\kappa_2(A)$, for example on poorly scaled matrices.

Make it small by construction

Run Jacobi not on A but on a **preconditioned** matrix $\tilde{A} = A\tilde{V}$ whose columns are nearly orthogonal. Get $\tilde{V}$ cheaply from a *low-precision* SVD; apply it in *high* precision so the product itself is accurate.

Three precisions are used, with $u_\ell > u > u_h$:

- ▶ low precision computes the preconditioner;
- ▶ high precision applies it, in one matrix–matrix product;
- ▶ working precision runs the one-sided Jacobi sweeps.

We need $\tilde{V}$ numerically orthogonal, $\|\tilde{V}^T \tilde{V} - I\|_2 = O(u)$, so that A and $\tilde{A}$ share singular values to within $O(u)$.

The algorithm

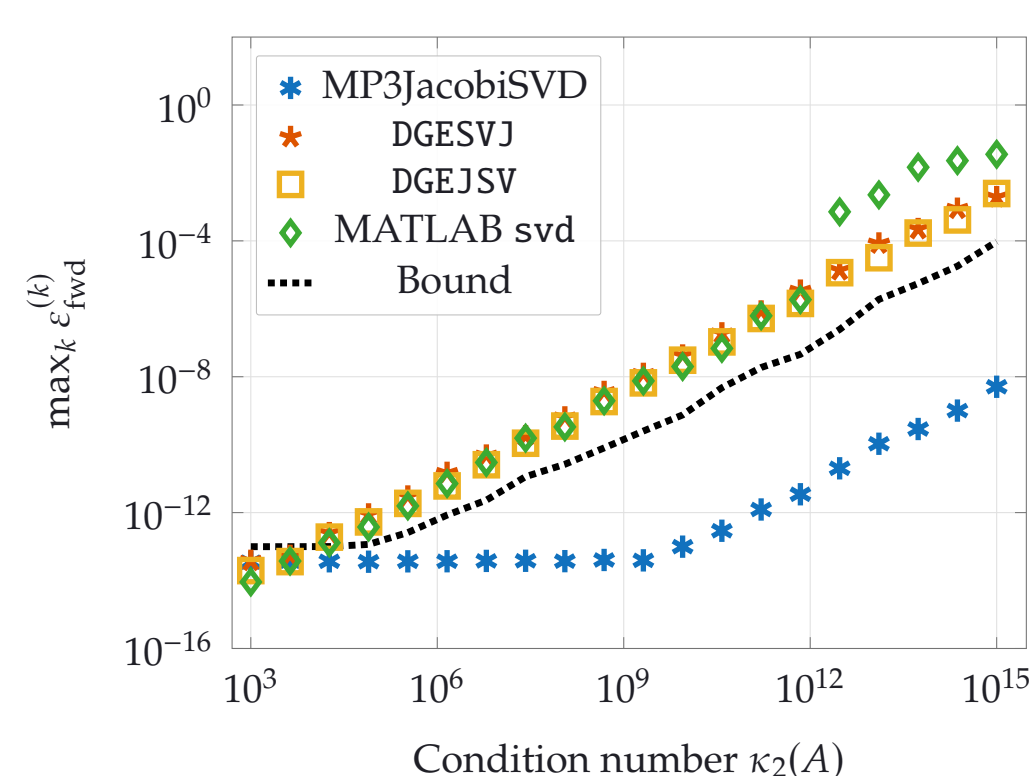
MP3JacobiSVD

Input $A \in \mathbb{R}^{m \times n}$, precisions $u_\ell > u > u_h$.

1. Right singular vectors V_ℓ of A at precision u_ℓ .
2. Householder-QR V_ℓ at precision u ; take the orthogonal factor as $\tilde{V}$.
3. Form $\tilde{A} = A\tilde{V}$ at precision u_h , round to u .
4. For tall-and-skinny matrices: reduced QR $\tilde{A} = Q_{\tilde{A}} R_{\tilde{A}}$ and run one-sided Jacobi on $R_{\tilde{A}}$; otherwise, run one-sided Jacobi on $\tilde{A}$.
5. Recover singular vectors using $\tilde{V}$, and optionally $Q_{\tilde{A}}$.

The QR factorization must come **after** the preconditioning. Before it, $\kappa_2^S(A)$ re-enters the error bound and accuracy is lost; after it, $R_{\tilde{A}}$ still has nearly orthogonal columns and the Jacobi phase still converges faster.

Singular values in practice



$A \in \mathbb{R}^{1000 \times 800}$ from `gallery('randsvd', [1000, 800], kappa, 5)`, so the singular values are random with uniformly distributed logarithms, at twenty logarithmically spaced $\kappa_2(A)$ from 10^3 to 10^{15} . Dotted line: the bound $(mn)^{1/2} u \kappa_2^S(\tilde{A})$. At $\kappa_2(A) \approx 10^{14}$, MP3JacobiSVD returns about **nine correct digits** where the other three return two or three.

Theorem: the singular values

[ZTW26a] Let $A \in \mathbb{R}^{m \times n}$ have full rank and $\tilde{A} = A\tilde{V}$ be as above. Under mild assumptions on the precisions, the computed k th singular value satisfies

$$\varepsilon_{\text{fwd}}^{(k)} \leq p(m, n)u + p(m, n)\kappa_2^S(\tilde{A})u.$$

The first term reflects that the preconditioner is only numerically orthogonal, the second the error from applying the Jacobi algorithm. This is the classical one-sided Jacobi bound with $\kappa_2^S(A)$ replaced by $\kappa_2^S(\tilde{A})$.

Obliquity: why $\kappa_2^S(\tilde{A})$ is small

We can use the following definition to show when $\kappa_2^S(\tilde{A})$ is small.

Definition (obliquity). For $A \in \mathbb{R}^{m \times n}$, $m \geq n$,

$$\text{obliq}(A) := \|AD - W\|_2,$$

with $D = \text{diag}(\|a_j\|_2^{-1})$ and W an orthogonal polar factor of AD .

It measures how far from orthogonal the columns are, and $\text{obliq}(AV_A) = 0$ for the exact right singular vectors.

- ▶ $\text{obliq}(\tilde{A}) \leq \theta < 1 \implies \kappa_2^S(\tilde{A}) \leq (1 + \theta)/(1 - \theta)$.
- ▶ If $u_\ell \kappa_2(A) \ll 1$ then $\kappa_2^S(\tilde{A}) \approx 1$ whatever $\kappa_2^S(A)$ may be.

The last line is a recipe for choosing precisions:

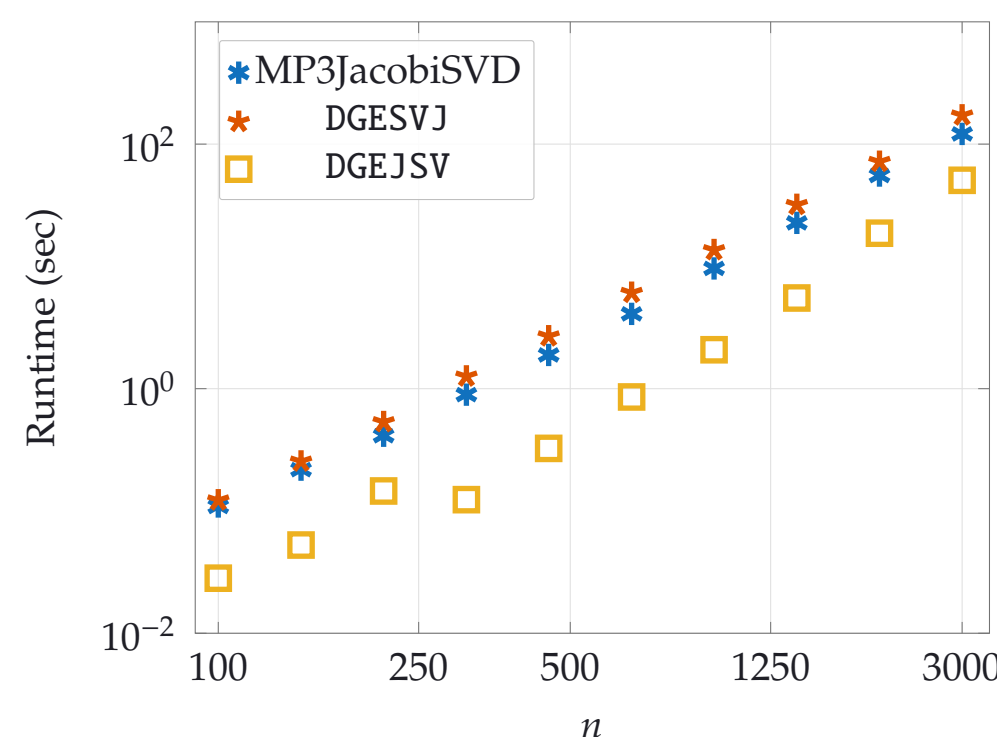
$$u_\ell \approx \kappa_2(A)^{-1}.$$

Timing results

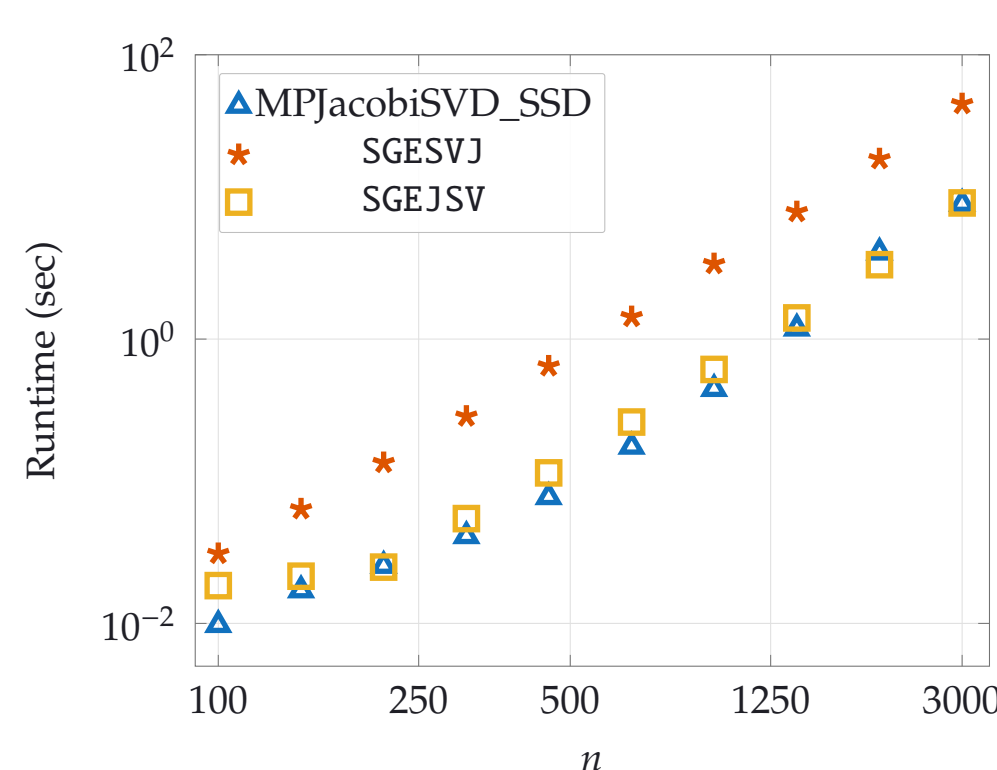
Profiling MP3JacobiSVD at (single, double, quadruple):

Quadruple-precision product $A\tilde{V}$	58.1 %
One-sided Jacobi (DGESVJ)	37.5 %

- ▶ **Faster than DGESVJ**, still behind DGEJSV.
- ▶ The bottleneck is simulated quadruple arithmetic: applying the preconditioner costs 230–490× what it costs in double. Optimized implementations get below 50×, and in theory about 7×.



Total runtime on $A \in \mathbb{R}^{3000 \times n}$ with $\kappa_2(A) = 10^8$ and geometrically distributed singular values.



The same algorithm at (single, single, double), every precision native on the hardware, against the single-precision LAPACK routines. As above but with $\kappa_2(A) = 10^6$.

Theorem: the singular vectors

Measure the error by the angle $\varepsilon_{\text{sv}}^{(k)} = \sin \angle(v_k, \widehat{v}_k)$.

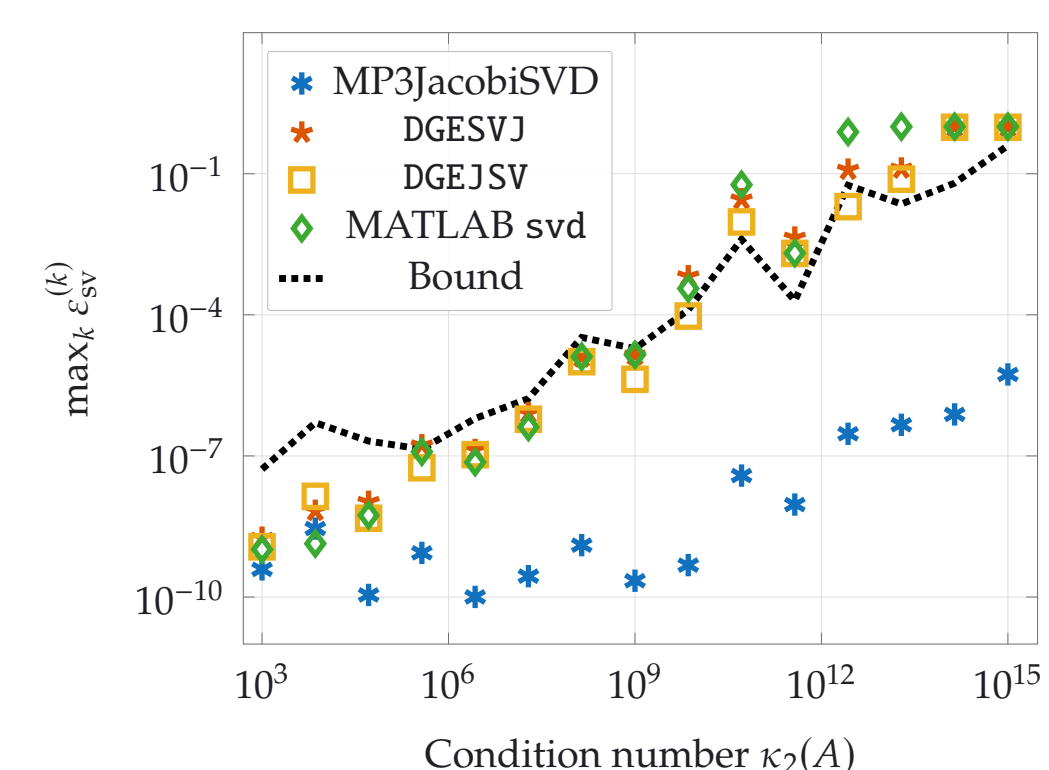
[ZTW26b] Under similar assumptions,

$$\varepsilon_{\text{sv}}^{(k)} \leq p_4 u \left(1 + \frac{\kappa_2^S(\tilde{A})}{\text{svrg}(A, k)} \right),$$

$$\text{svrg}(A, k) = \min \left\{ 2, \min_{j \neq k} \frac{|\sigma_k - \sigma_j|}{\sigma_k} \right\}.$$

It holds for the left singular vectors too.

- ▶ The **relative** gap appears, not the absolute gap. Bidiagonalization-based methods divide by $\min_{j \neq k} |\sigma_k - \sigma_j|$, which is necessarily tiny once two singular values are small.
- ▶ Again $\kappa_2^S(\tilde{A})$, not $\kappa_2^S(A)$.
- ▶ So the gain is largest for ill-conditioned matrices with *small absolute but moderate relative* gaps.



Largest $\sin \angle(v_k, \widehat{v}_k)$ over all right singular vectors of $A \in \mathbb{R}^{1000 \times 800}$, generated the same way, at fifteen logarithmically spaced $\kappa_2(A)$ from 10^3 to 10^{15} . Dotted line: the bound $(mn)^{1/2} u (1 + \kappa_2^S(\tilde{A})/\text{svrg}(A, k_*))$ at the index k_* that attains the error.

Takeaways and open problems

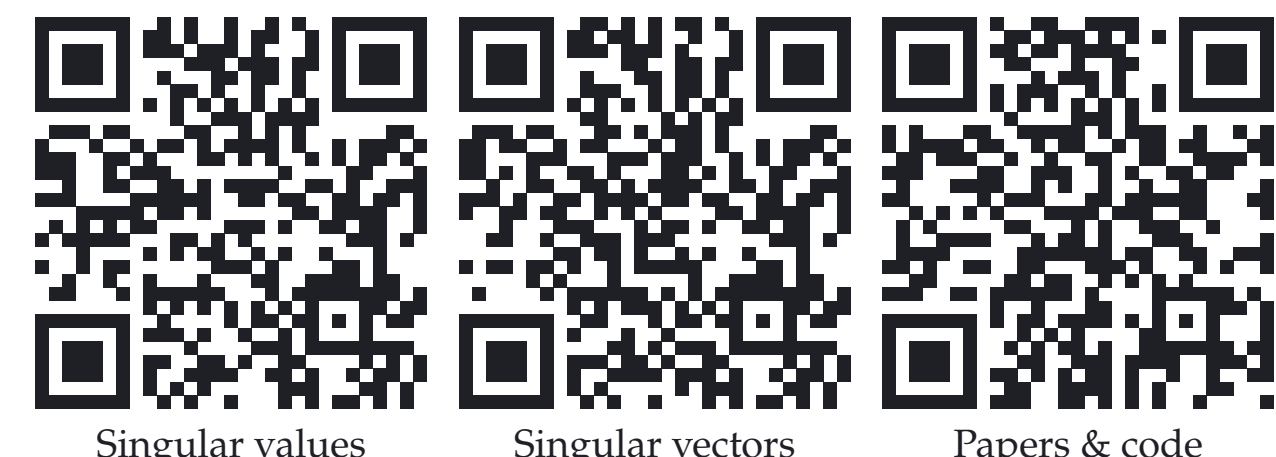
A highly accurate SVD is achieved by:

- ▶ one low-precision SVD for constructing the preconditioner, and
 - ▶ one high-precision matrix product for preconditioning.
- Performance is competitive with DGEJSV if high-precision preconditioning is fast.

- ▶ Improved implementations of high-precision matrix product can directly make MP3JacobiSVD faster.
- ▶ A production implementation would also need scaling and overflow handling, which DGESVJ and DGEJSV already carry.
- ▶ Clustered and multiple singular values are open.
- ▶ Experiments show the complex case behaves identically; the analysis is not yet written.

References

- [DV92] James Demmel and Krešimir Veselić. “Jacobi’s method is more accurate than QR”. In: *SIAM J. Matrix Anal. Appl.* 13.4 (Oct. 1992), pp. 1204–1245.
- [ZTW26a] Zhengbo Zhou, Françoise Tisseur, and Marcus Webb. *Computing accurate singular values using a mixed-precision one-sided Jacobi algorithm*. arXiv:2602.18134. Feb. 2026.
- [ZTW26b] Zhengbo Zhou, Françoise Tisseur, and Marcus Webb. *Computing accurate singular vectors and eigenvectors using mixed-precision Jacobi algorithms*. arXiv:2606.28069. June 2026.



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